Computers, Cognitivists and Controversy

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The cognitive approach to psychology was a departure from the previous traditions of psychoanalysis and behaviourism — arising out of the development of sophisticated World War II technology as well the popularization of the digital computer in the late 1960s, it aimed to study how the mind and mental processes "transformed, reduced, elaborated, stored, recovered, and used" sensory input (Neisser, 1967, p. 4). However, recent discourse around the topic of advanced technology has called into question the cognitive tendency to model the mind on computers and describe it using mechanical terms. In this essay, I posit that the foundations of the field of cognitive psychology have become increasingly controversial as a consequence of the rising threat of artificial intelligence (AI) replacing human thought and behaviour.

The term 'artificial intelligence' was coined by Alan Newell and Herbert Simon in 1956 when they presented their automated reasoning program, Logic Theorist (Russell & Norvig, 2020). The very same year, Miller's (1956) seminal presentation 'The Magic Number Seven' not only brought into psychological view the then-developing of information theory, but also provided a precedent for applied sciences to inform psychological frameworks. A burgeoning penchant for computational connectionist terms can be seen as early as Chomsky's fatal critique on behaviourism, which has been widely seen as the lynchpin for the cognitive revolution (Hergenhahn & Henley, 2014): he calls for clarification on "the actual units of verbal behaviour" (Note 46, Chomsky, 1959). Chomsky's linguistic position is particularly salient: the interaction between linguistics, psychology and neurology as they confronted the rise of technology culminated in what we now call the inter-disciplinary 'cognitive science' approach (Hatfield, 2002). The subsequent rise of cognitive science, paralleled with the advent of the digital computer, thus ostensibly and irrevocably linked the oeuvre of cognitive psychology with technology.

The force of the cognitive revolution was, indeed, pivoted on machines that applied parallel processing: prior to the 1970s, computers could only handle data one at a time through serial processing, evidently nothing like the human mind (Neisser, 1976 as in Tschacher & Scheier, 2012). Cognitivists

regarded the mind as analogous to the computer by virtue of the fact that such computer could execute functions similar to humans. By explicitly citing the computer as a metaphor for the mind, the mind was conceptualized as a complex information-handling machine (Green, 1967 as in Hatfield, 2002). The parallel processing computers were the prototype for some of the most popular models of cognition, called 'connectionist' models of the mind. In such models, the mind was made up of 'units' that were connected to each other by networks that provided feedback mechanisms (Seidenberg, 2005; see Figure 1) – an evidently computational approach to thinking.

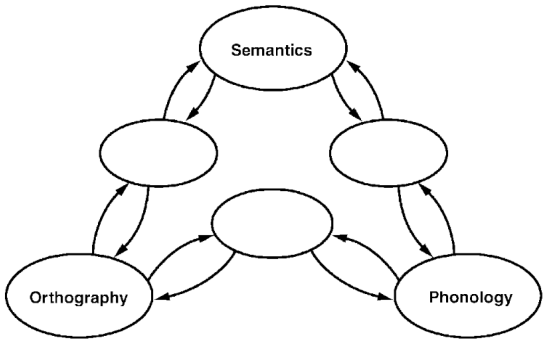


Fig. 1. Theoretical framework (introduced by Seidenberg and McClelland, 1989) that has served as the basis for several implemented models of word reading. Collectively known as the ''triangle model,'' the models differ in details and focus but are based on the same theoretical principles. Large ovals represent groups (''layers'') of units that encode different types of information: orthography (spelling), phonology (derived from pronunciation and sound), and semantics (meaning). Smaller ovals represent ''hidden units,'' which increase the computational capacity of the network and provide the basis for abstraction. Most models have focused on the orthography-to-phonology mapping. Harm and Seidenberg (2004) implemented both orthography–semantics and orthography–phonology–semantics components, using a variant of this architecture.

Figure 1. The connectionist model for word processing, as seen in Seidenberg, 2005.

Not only did the cognitivists use the computer metaphor to create theoretical frameworks, they began to use computers as evidence for their models. For example, Seidenberg (2005) cited a computer program that implemented and successfully simulated his model of human learning in word-recognition (see Figure 1) as strong evidence that the model was representative of the human mind. It is easy to see a sort of circular reasoning that emerges with these 'evidentiary' models in the cognitive view: it is not the mind (a natural subject) that is analogous to the computer (the man-made object), but the other way around.

Even in its heyday, despite initial success and widespread acceptance, there were similar concerns

about the limits of a computational metaphor for the mind. Neisser himself, in the introduction to what would become the handbook for cognitive psychology, warned against reducing human mental processes to that of a computer, asserting that "the rest of this book can be considered an extensive argument against models of this kind" (1967, p. 9). Further discourse revealed criticism over the dangerous assumption that all minds were "systematic" in their mental representations (Casey & Moran, 1989; Fodor & Pylyshyn, 1988). Outside of psychology, other disciplines recognised this failure: sociologist Alan Wolfe famously wrote about the cognitive approach to psychology that "in substituting procedures for meaning, they again fail to account for how human brains, let alone human minds, work" (1991). Frank Jackson, through his landmark thought-experiment (1986) 'Mary's Room', illustrated that only the human mind could experience and simulate qualities of the world. He asks the readers to imagine a woman named Mary who has grown up in a black-and-white room with all the information about the outside world provided to her, and then asks a simple question: will Mary be surprised when she leaves her room? The intuitive answer is yes. This thought-experiment undoes the cognitivist perspective succinctly — the information in the room should have given her a means (or a set of rules similar to Seidenberg's (2005) computer program above) to manipulate the sensory input she will receive outside; the fact that a human mind intuitively learns through qualia, or something other than pure information, posed a significant problem for the cognitivists' evidentiary computer models.

The proponents of cognitive psychology attempted to defend their framework against these criticisms by emphasizing layers of representation (Bechtel & Herschbach, 2010) — the cognitive models had previously rested on explanation of behaviour as a result of bits of information as symbolic representations in the mind undergoing different processes; now, they were required to identify the semantic content of these symbols. This conundrum surrounding semantics was thrust into mass spheres and became a serious problem for cognitivists at the turn of the century with John Searle's (2002) famous 'Chinese Room' thought experiment, which argued that a human mind was distinct from a computer based purely on its ability to generate meaning. He proposes a scenario where someone who speaks no

Chinese attempts to communicate with someone who is a native Chinese speaker by referring to dictionaries and grammatical guides. Despite being able to successfully communicate, there is no case in which one could claim that the non-Chinese speaker understands Chinese — he was simply following the rules in the books. Here, the non-Chinese speaker is exemplary of the cognitive model: able to simulate human behaviour but unable to handle any semantic explanation of it. Searle's attack on the hypothetical mental operations that act on representations in the cognitive perspective hence affirmed that these operations *have* to operate differentially on the symbolic and semantic aspects of a representation to have any bearing on the human mind.

Intellectuals thus began to assert that the human mind was distinct in its ability to go beyond symbols, syntactical forms and constrained rules that a computer programme follows in order to attach *meaning* to our experiences. Cognitive psychologists, in turn, amended their models — they focused on 'modular' systems that explained that a presentation in the mind could differentially affect behaviours based on which module in the mind it activated, and the basis of that difference was semantic meaning (Samuels 1998 as in Machery, 2010). Other than the unclear definition of what constitutes a module, they may have well succeeded at their defence, if not for the unprecedented strides in the advancement of computers and technology. By the early 2000s, the abilities of a computer that had first met the abilities of the average human now far surpassed them: at present, tools like ChatGPT and Adobe's Generative Fill are now able to consult vast amounts of data, process it, and generate original text and images a hundred times faster than *any* humans.

The rise of these tools and programmes, collectively held under the umbrella of AI, has garnered much concern among intellectuals about not only the position, but also the nature of human thought, creativity, and function. It has called for reflection across multiple disciplines (but especially in philosophy, literature and art) on what it means to be fundamentally human — and this shift is not purely academic, it can be clearly reflected in all forms of popular media, from TV shows like 'Black Mirror', movies like 'Don't Worry Darling' and podcasts like 'The Magnus Archives', which display a general

distrust against AI and robots. Under these circumstances, cognitive psychology must reconcile and defend their models of the mind that situate consciousness in a metaphorical computer. Cognitive psychology has become a site of investigation that "[…] needed foundations, which it would be a legitimate job for philosophy to provide" (Thagard, 2009, p. 251). In other words, the subject of the mind has become so inextricable from wider narratives that any new theoretical frameworks or empirical findings in cognitive science would by default affect its related fields of philosophy, neurology and computer science, and vice-versa.

In essence, the perspective of cognitivists, whose theories and research focused on uncovering the mind in terms of 'modules', 'domains', 'networks', 'automaticity', and 'connections' has been called into serious question. Cognitivists have come up against the boundaries of their own field in this search, acknowledging that a " […] scientific understanding of intentionality is still lacking". (Tschacher & Scheier, 2012, p. 70). The comparison of the human mind to a computer has become a gross abstraction that reveals very little about the function of the mind outside of broad models. This abstraction occurs at "the 'software' or 'program' level of description, rather than at the level of physical implementation", completely eschewing any material systems (Bechtel & Herschbach, 2010). However, all these charges against cognitivism cumulatively escalate in the face of the threat of AI – the need to distinguish between human mind and artificial mind have resulted in an impasse that makes the computer metaphor seem unfathomable in the current climate.

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